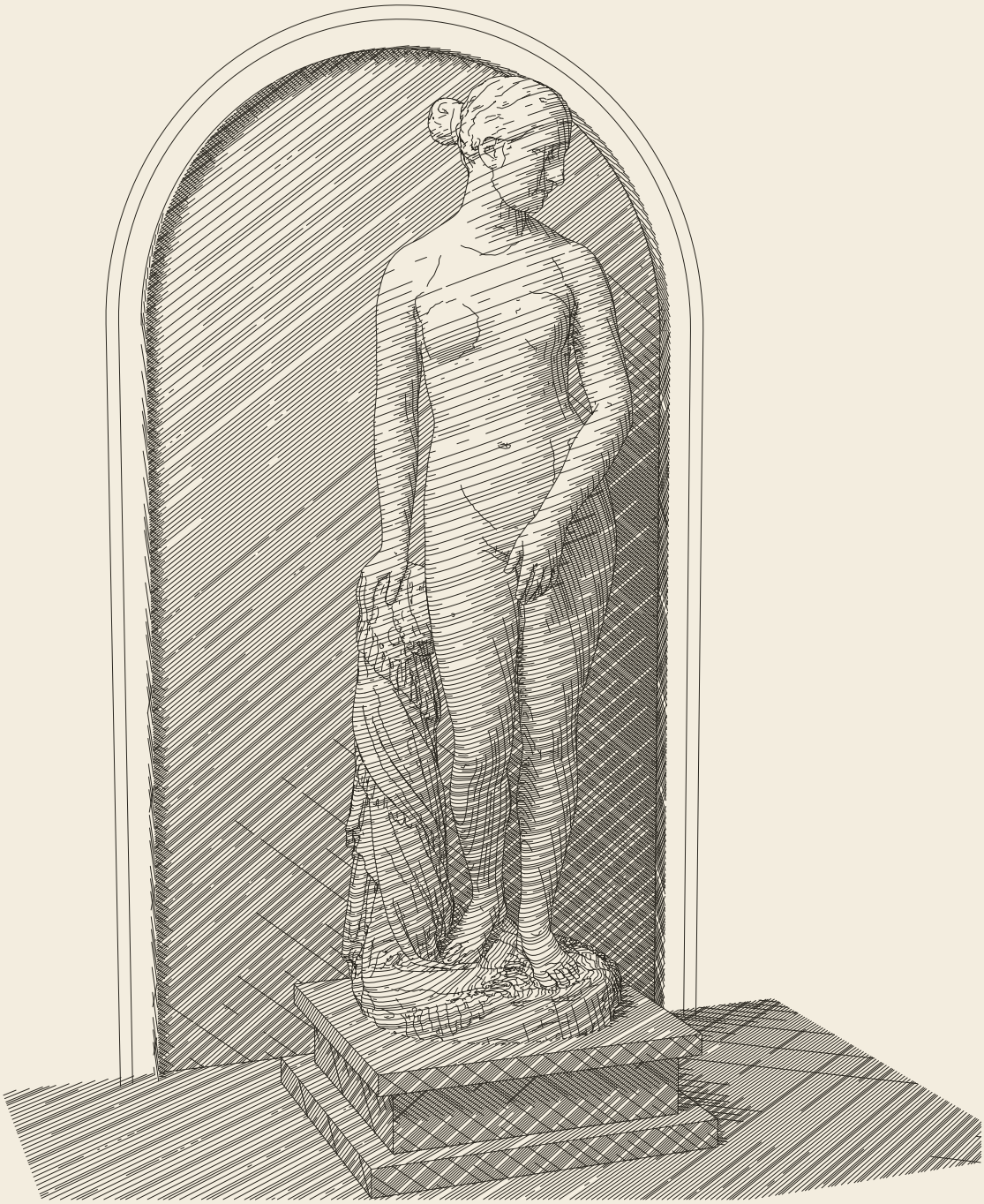
I. A figure in the recess



Contours, fold lines and surface hatches. 39.1 m of line at 148 mm.

Hiram Powers, Model of the Greek Slave (1843). Smithsonian American Art Museum. Scan CC0.

II. The second angle



The camera of the Trail First study. 48.7 m of line at 148 mm.

Same scan, light, setting and tone curve as v1; geometry and light are read at stroke resolution.

Where the detail was going



v1, 148 mm



v2, 148 mm



v2, 296 mm, same pen

The head, enlarged 4x from the plate (about 18 x 19 mm at 148 mm).

Outlines. The GLB is split along its texture atlas: 94,531 vertex records for 74,939 positions. v1 found silhouettes by vertex index, so they broke at every seam: 4,101 chains with a median of 3 vertices, drawn as dashes. v2 welds the seams and draws the zero set of n.v on vertex normals: 430 continuous chains, tested for visibility from a point lifted just off the surface.

Fold lines. Suggestive contours (DeCarlo et al., 2003) draw the eyelids, the side of the nose, the line of the lips and the jaw. On a lit face, tone alone cannot carry these at this scale.

Light. v1 read light from a 600 x 760 screen buffer (2.5 plate units per sample, flat triangle normals, 3 x 3 median). v2 shades the scan's vertices with 64 light, 32 occlusion and 12 bounce samples, and the setting on 0.01-unit texture grids, then interpolates to each stroke sample. The tone curve is unchanged.

Line spacing. v1 spaced candidates 0.008 scene units apart, about 3.1 pen widths, at any plate size. v2 spaces them 1.75 pen widths apart on paper. At 148 mm that gives 39 m of line. At 296 mm with the same 0.09 mm pen it gives 150 m, which is where the lit face starts to carry tone.

Not changed: scan, pose, camera, light position, exposure, tone curve, hatch plane families and the nested on/off order (now 64 steps instead of 32). The 150k-triangle mesh projects to ~3.9 plate units per edge, so at 296 mm about five hatch lines cross each triangle. Beyond that, the museum's larger download becomes the limit.

Trail First at plate resolution



v1, unchanged sampler



Unchanged rules, 0.12 mm grid



Guided food, marks on trails

The vendor sampler works in grid cells: sensors 6-8 cells, steps of 1.25 cells, 3 x 3 diffusion. v1 fixed the grid at 192 cells across. One cell was 7.8 plate units on the full frame and 2.45 on the crop. On the full frame the sensors reached 47-62 plate units, about a third of the head's height, and the trails merged into one trunk.

Here a cell is 1.25 plate units (0.12 mm), with 10,450 agents per colony scaled to the statue's area. The vendor code is unchanged. Under the original rules the network now branches inside the figure. With the guided food (darkness plus gradient), the trails follow edges. Placing the marks on the trails rather than the food draws the figure as a network: 20,000 marks, 0.08-0.2 mm.

The mark budget is the other half of resolution. 7,200 marks over the figure is about one per 45 square plate units; in v1 about 450 of them landed in the whole head and neck.